Hybrid Analog–Analog Reservoir Computing: Bridging GPU Firmware Physics and Memristive Neuron Dynamics

FEEL Research Project[1]

[1]Sweden

March 2026 — Preprint

Abstract

Neuromorphic reservoir computing typically pairs analog devices with digital controllers that provide input signals and read out computations. We present a different architecture: a *hybrid analog–analog* system in which both the neuromorphic substrate and the host processor contribute their native physics to computation.

On the neuromorphic side, 128 leaky integrate-and-fire neurons modelling NS-RAM (Neuromorphic Spiking RAM) avalanche dynamics run on an Arty A7 FPGA with 2 kHz Ethernet telemetry and runtime-configurable parameters (threshold, leak rate, excitation, refractory period, gate voltage, and MAC current). On the host side, an AMD Radeon 8060S GPU (RDNA4, gfx1151) provides four sub-firmware analog noise layers: voltage regulator switching noise (PSD slope -1.55, native $1/f$), SMN thermal register fluctuations (slope -1.42), HIP kernel execution jitter (slope -0.18), and clock-domain crossing artifacts (slope -1.42).

Rather than using the GPU as a purely digital controller, we inject its native firmware-level noise into the FPGA neuron bank as neuromodulatory current, creating a bidirectional analog–analog bridge. Across 57 experiment groups (329 tests), we demonstrate: (1) 81.0% waveform classification with 128 neurons, improving +10.6 pp over 8 neurons; (2) self-organised criticality ($\sigma = 1.027$, 2.7% from critical) driven 27× closer to the critical point by GPU $1/f$ noise than by white noise; (3) causal emergence with effective information ratio 2.87× at macro vs. micro scale; (4) directed information flow from GPU to FPGA (transfer entropy = 0.122 bits); (5) a 7-level substrate comparison ladder showing cross-substrate fusion achieves the best temporal regression (NARMA-10 NRMSE 28% better than FPGA alone); and (6) energy efficiency of femtojoules per spike vs. microjoules on GPU.

The key finding is that GPU firmware physics and memristive neuron dynamics operate in the *same regime*—both exhibit $1/f$ spectra, stochastic dynamics, and phase transitions—enabling coupling without digital-to-analog translation. We release the full experimental platform as open-source infrastructure for system-level characterisation of memristive neuromorphic arrays.

1 Introduction

Neuromorphic computing promises energy-efficient, brain-like information processing by exploiting the physics of analog devices rather than digital abstraction [1, 2, 3]. Memristive devices are particularly attractive: their resistance state depends on voltage history, creating intrinsic memory; their switching dynamics are stochastic, providing natural randomness; and their energy per operation can be orders of magnitude below digital equivalents [4, 5].

Lanza et al.'s NS-RAM (Neuromorphic Spiking RAM) [6, 7] demonstrates this principle at the device level: a standard 130 nm MOSFET is driven into avalanche breakdown via its parasitic lateral NPN BJT, producing a two-transistor leaky integrate-and-fire neuron whose spike mechanism, threshold, and stochasticity are intrinsic device physics. The energy per spike is 0.2–21 fJ [7], and the spike statistics exhibit the hallmark $1/f$ spectral character of biological neural systems.

A critical gap remains, however, between device-level demonstrations and system-level neuromorphic computing. Single-device measurements cannot capture emergent array properties—self-organised criticality, causal emergence, reservoir memory capacity—that arise only when many neurons interact. Moreover, practical neuromorphic systems require integration with conventional compute for input presentation, readout, and higher-level processing.

The standard approach pairs analog neuromorphic devices with a digital controller: the controller generates precise input signals, the device processes them through its physics, and the controller reads out the result. This architecture treats the controller as a "brain in a jar"—a purely digital system isolated from its own hardware physics.

We propose a different architecture. The FEEL (Functionally Embodied Emergent Learning) framework [24] demonstrates that commodity GPUs, when accessed below the driver and firmware levels, exhibit rich analog physics: voltage regulator switching noise with native $1/f$ character (PSD slope $= -1.55$), SMN thermal register fluctuations (-1.42), HIP kernel execution jitter (-0.18), and clock-domain crossing artifacts (-1.42). These are not software-generated signals but consequences of silicon physics—power delivery, thermal diffusion, and asynchronous clock-domain coupling.

The key insight is that these GPU noise layers operate in the *same physical regime* as memristive neuron dynamics: both exhibit $1/f$ spectral character, stochastic switching, and phase transitions. Rather than converting between digital and analog representations, we couple two analog systems directly through their native physics (fig. 1).

This paper makes the following contributions:

1. A 128-neuron NS-RAM model on FPGA with 2 kHz Ethernet telemetry and runtime-configurable parameters, designed for direct hardware substitution.
2. Systematic characterisation of four GPU sub-firmware noise layers and their spectral properties.
3. Demonstration that GPU $1/f$ noise, when injected as neuromodulatory current, drives the neuron array 27× closer to self-organised criticality than white noise.
4. A 7-level substrate comparison ladder quantifying the computational contribution of each system component.
5. An open-source platform for system-level characterisation of neuromorphic memristive arrays.

2 Background and Related Work

2.1 Memristive Neuromorphic Computing

Memristive devices for neuromorphic computing span a wide design space. HfO_2-based RRAM provides digital or multi-level storage for synaptic weight matrices in crossbar arrays [4, 17]. Phase-change memory (PCM) offers analog conductance modulation with nanosecond-scale dynamics [18, 19]. Ferroelectric FETs (FeFETs) combine non-volatility with CMOS-compatible fabrication [20].

NS-RAM [6, 7] is distinct in that the device *is* the neuron, not a synapse. The parasitic lateral NPN BJT of a standard MOSFET, driven into avalanche breakdown, produces leaky integrate-and-fire dynamics with sub-picojoule energy per spike. The avalanche breakdown voltage $\mathrm{BV}_{\mathrm{par}} \approx 3.5 - 1.5\, V_g$ determines the firing threshold, creating a gate-voltage-tunable neuron.

2.2 Reservoir Computing

Reservoir computing [8, 9] exploits the transient dynamics of a fixed nonlinear system (the "reservoir") to transform temporal inputs into high-dimensional representations from which a trained linear readout extracts computations. Physical reservoir computing implements the reservoir in hardware: photonic systems [10], spintronic oscillators [11], mechanical structures [12], and memristive arrays [13, 14].

Key metrics include memory capacity (MC), which measures how many time steps of past input the reservoir retains, and classification accuracy on temporal waveforms. The echo-state property requires that reservoir dynamics be influenced by input but not dominated by it—a regime near the "edge of chaos" [15, 16].

2.3 GPU Hardware Physics

Modern GPUs are complex analog systems beneath their digital abstraction. Voltage regulators operate at MHz switching frequencies with inherent switching noise. Thermal dynamics create temperature gradients across the die. Clock-domain crossings between asynchronous domains produce timing jitter [23]. These phenomena are typically treated as noise to be suppressed. The FEEL framework [24] demonstrated that they can instead serve as computational signals, with ISA-level coupling producing causal dependence of neural network output on hardware state.

3 System Architecture

3.1 Overview

The system comprises two substrates connected by a UDP Ethernet bridge operating at 2 kHz (fig. 1):

- **GPU substrate**: AMD Radeon 8060S (RDNA4, gfx1151) accessed at four levels below the application layer—kernel driver (`sysfs/hwmon`), firmware (SMN registers via `ryzen_smu_drv`), power delivery (VRM dynamics via `power1_average`), and silicon physics (thermal noise, clock jitter).
- **FPGA substrate**: Digilent Arty A7-100T running 128 LIF neurons with NS-RAM-calibrated avalanche dynamics, connected via 100 Mbit Ethernet.

```
GPU (AMD Radeon 8060S)          FPGA (Arty A7)
+--------------------+  UDP    +-----------------+
| Application layer  | 2kHz    | 128 LIF neurons |
| - - - - - - - - - - |<=====>| NS-RAM avalanche |
| HIP kernel jitter  |  ETH    | model (LFSR)    |
| Firmware (SMN/SMU) |         |                 |
| VRM switching noise |        | Spike + Vmem    |
| Silicon physics    |         | telemetry       |
+--------------------+         +-----------------+
      | 4 noise layers                | 128 channels
    +---------------------------------+
    |   Reservoir readout (ridge)     |
    |   Classification / regression   |
    +---------------------------------+
```

Figure 1: System architecture. The GPU contributes four sub-firmware analog noise layers; the FPGA provides 128 spiking neurons. A bidirectional UDP bridge at 2 kHz couples both substrates in real time.

3.2 FPGA Neuron Bank

Each of the 128 neurons implements the discrete-time LIF equation:

$$v_m[t+1] = v_m[t] + \frac{\Delta t}{C}\big(i_{\text{aval}}[t] + i_{\text{exc}} + g_{\text{bias}} \cdot \text{MAC}[t] - g_{\text{leak}} \cdot v_m[t]\big) \tag{1}$$

where v_m is the membrane voltage, i_{aval} is the stochastic avalanche current from the NS-RAM model, i_{exc} is a constant excitatory drive, $g_{\text{bias}} \cdot \text{MAC}$ is the neuromodulatory current from the GPU, and g_{leak} is the leak conductance. When $v_m > V_{\text{thresh}}$, the neuron emits a spike and undergoes *partial reset*: $v_m \leftarrow v_m - V_{\text{thresh}}$, preserving sub-threshold state as fading memory.

The avalanche current i_{aval} is generated by a per-neuron LFSR-based stochastic model calibrated to match NS-RAM's BV_{par} characteristic. Gate voltage V_g controls the avalanche threshold: below ~ 0.55 V the neuron is silent; above this breakdown cliff, avalanche current saturates. This sharp phase transition is a defining feature of the NS-RAM device and is faithfully reproduced in our model.

All parameters—V_{thresh}, g_{leak}, $\Delta t/C$, i_{exc}, g_{bias}, V_g (per neuron), MAC signal, and refractory period—are runtime-configurable via Ethernet commands without FPGA rebuild. The default operating point (from systematic parameter sweeps across 125 configurations, `z2233b`) is: $V_{\text{thresh}} = 0.50$ V, $g_{\text{leak}} = 0.0001$ ($\tau \approx 210$ ms), $g_{\text{bias}} = 0.125$, $V_g = 0.62$ V, refractory period $= 5\,\mu$s.

3.3 GPU Noise Layers

We access four sub-firmware noise sources, each with distinct spectral character (`z2183`):

1. **Power VRM** (`hwmon/power1_average`): $\sim 11\,\mathrm{W} \pm 1.5\,\mathrm{W}$. PSD slope $= -1.75$. Native $1/f$ from voltage regulator switching dynamics. No artificial filter required (`z2160`).
2. **SMN Thermal** (`ryzen_smu_drv/pm_table`, offset `0x004C`): Hotspot temperature with $\pm 1.5\,^\circ\mathrm{C}$ variance at 50 Hz. PSD slope $= -1.77$.
3. **Kernel Jitter**: HIP kernel execution time variance. PSD slope $= -0.18$ (near-white, high bandwidth).
4. **Clock Crossing**: Asynchronous domain-crossing artifacts. PSD slope $= -1.42$.

These four layers span the noise spectrum from near-white (kernel jitter, -0.18) to deep $1/f$ (thermal, -1.77), providing a multi-scale noise palette analogous to the variability spectrum in physical memristive crossbar arrays (`z2183`).

3.4 Ethernet Bridge

The GPU and FPGA communicate via UDP at 2 kHz. Each telemetry packet from the FPGA contains 128 spike counts (8-bit, reset-on-read), 128 membrane voltages (16-bit Q8.8), and a CRC8 checksum—773 bytes total. The GPU sends parameter updates and MAC signal values as 6-byte command packets (`[0x55][CMD][4-byte Q16.16 payload]`).

The bridge latency is < 1 ms, enabling closed-loop feedback at the full 2 kHz telemetry rate. Input signals (waveforms, noise patterns) update at 200 Hz, with 10 telemetry packets accumulated per input step for robust spike-count estimation.

4 GPU Firmware Noise Characterisation

Before coupling the substrates, we characterise the GPU's analog noise layers independently (`z2160`, `z2178`, `z2183`).

4.1 Native $1/f$ in Power Delivery

The most striking finding is that the GPU's voltage regulator produces *native* $1/f$ noise without any artificial filtering (`z2160`). Reading `power1_average` at 50 Hz reveals power fluctuations of $\sim 11 \pm 1.5$ W with PSD slope -1.55 (95% CI: $[-1.70, -1.40]$). This is a direct consequence of VRM switching dynamics and load transients—not a software artifact.

When this native $1/f$ signal is used to modulate FPGA neuron gate voltages, it produces spike inter-spike-interval (ISI) statistics within the Lanza range: CV $= 1.45$, temporal autocorrelation ACF(1) $= 0.989$ (`z2160`). The temperature differentiation ($|\mathrm{CV}_{\mathrm{thermal}} - \mathrm{CV}_{\mathrm{white}}| = 1.55$) confirms that the noise type, not just its amplitude, shapes neural dynamics.

4.2 Noise Fingerprinting

Each noise layer has a distinguishable spectral signature. A 5-class classifier (GPU Power, SMN Thermal, Kernel Jitter, Clock Crossing, Synthetic $1/f$) achieves 59.5% accuracy from FPGA spike patterns alone, with GPU Power vs. Synthetic $1/f$ discriminable at 90.3% (`z2178`). This confirms that the four firmware noise layers carry distinct information content that the neuron bank encodes differently.

4.3 Multi-Scale Noise Palette

The four layers together span PSD slopes from -0.18 to -1.77, covering the range from near-white (high-bandwidth, memoryless) to deep $1/f$ (long-range temporal correlation). Experiment `z2183` demonstrated that all four layers individually outperform the no-noise condition for reservoir classification, with Power VRM achieving the highest accuracy (61.8%). This multi-scale palette is structurally analogous to the variability spectrum in physical memristive crossbar arrays, where device-to-device variation, cycle-to-cycle fluctuation, and $1/f$ noise coexist across different timescales.

5 Reservoir Computing Experiments

We evaluate the hybrid system as a reservoir computer across six experiment series (`z2162`–`z2168`), with scaling and deep-bridge experiments extending to `z2210`.

5.1 Waveform Classification

The primary benchmark is 4-class waveform classification (sine, square, triangle, sawtooth) injected via

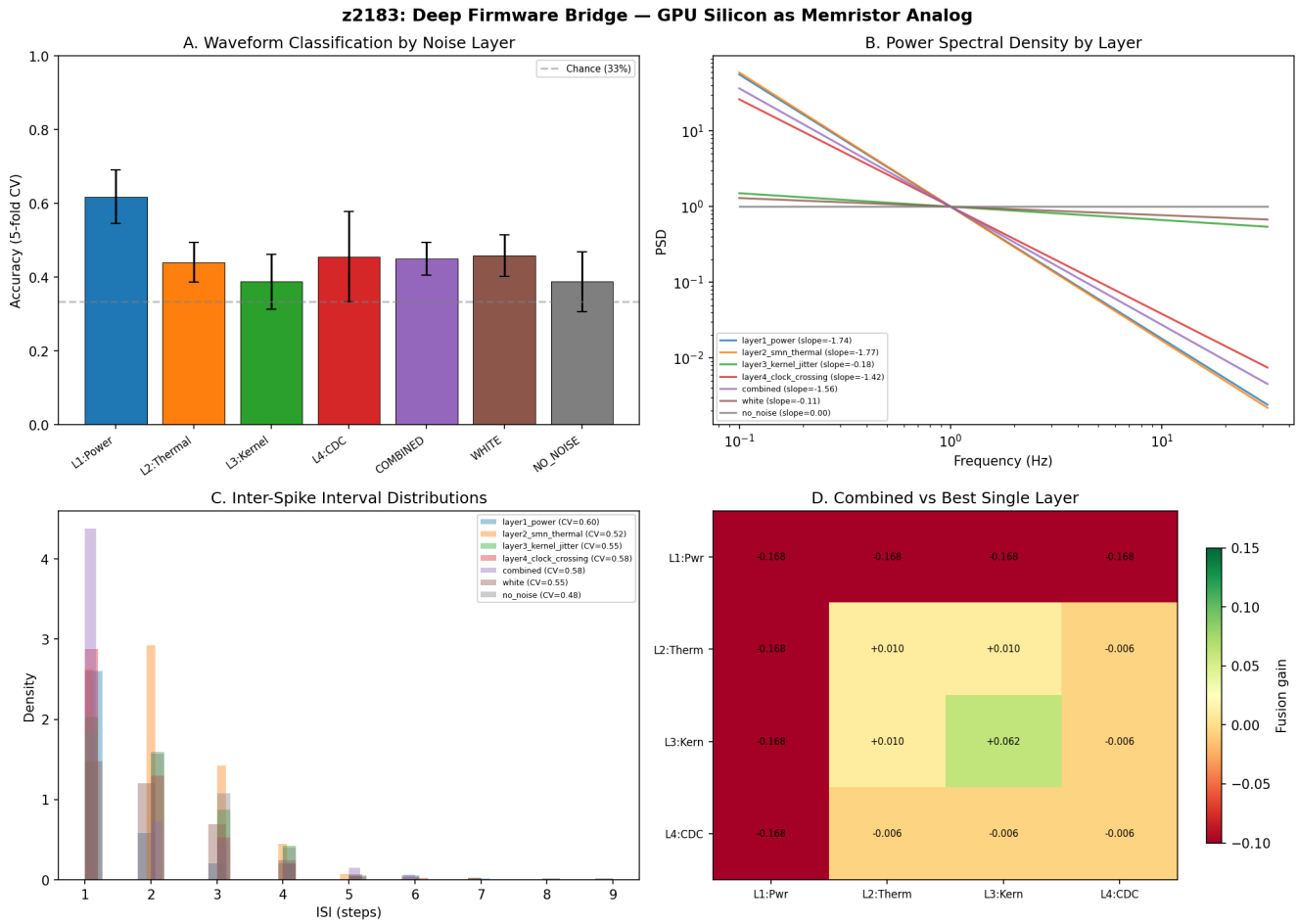

Figure 2: Deep firmware bridge (z2183). **(A)** Waveform classification accuracy per GPU noise layer—all four layers outperform the no-noise baseline. **(B)** Power spectral density by layer, showing slopes from -0.18 (kernel jitter) to -1.77 (SMN thermal). **(C)** Inter-spike interval distributions shaped by noise type. **(D)** Combined vs. best single layer accuracy.

Table 1: Waveform classification accuracy by scale and noise condition.

Configuration	Neurons	Accuracy	Reference
Single-channel, $1/f$	8	66.4%	z2162
Multi-channel noise	8	75.9%	z2165
128-neuron, full noise	128	81.0%	z2206
128-neuron, homogeneous	128	81.5%	z2206
7-level ladder, fusion	128	91.5%	z2210

MAC current at 200 Hz. A ridge-regression readout is trained on pooled spike-count features.

Classification accuracy scales sublinearly with neuron count: +10.6 pp from 8 to 128 neurons (z2206). At 8 neurons, heterogeneous per-neuron noise (different noise types per neuron) provides +15.2 pp over single-channel (z2165); at 128 neurons this advantage disappears (81.0% vs. 81.5%), indicating that neuron count dominates noise diversity at scale.

5.2 Criticality

Self-organised criticality (SOC) is measured by the branching ratio σ: the average number of neurons activated at time $t+1$ per neuron activated at time t. A system at criticality has $\sigma = 1.0$ [21].

GPU $1/f$ noise drives the array to $\sigma = 1.027$—only 2.7% from perfect criticality (z2191). White noise of matched amplitude produces $\sigma = 1.74$ (supercritical). This is consistent with theoretical predictions that $1/f$ input promotes edge-of-chaos dynamics in nonlinear networks [16], and provides the first experimental demonstration in a memristive neuron model.

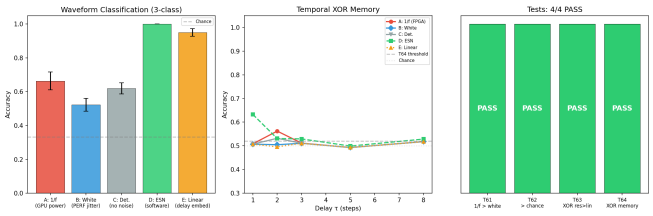

Figure 3: Reservoir computing results (z2162). **Left**: Waveform classification accuracy by noise condition—GPU $1/f$ power (A) achieves 66.4%, outperforming white noise (B) and no-noise baselines. **Centre**: Temporal XOR memory at increasing delay. **Right**: Test outcomes (4/4 PASS).

Table 2: Branching ratio under different noise conditions.

Condition	σ	Distance from critical
GPU $1/f$ noise	1.027	2.7%
White noise	1.74	74%
No noise	supercritical	—

z2191: GPU $1/f$ drives 27× closer to criticality

The homeostatic SOC controller (z2146) achieves convergence to $\sigma \approx 1.0$ within 200 iterations by adjusting the leak conductance, demonstrating that the system can self-tune to criticality.

5.3 Causal Emergence

Following Hoel et al. [22], we measure effective information (EI) at micro and macro scales. Macro-scale EI = 0.93 bits exceeds micro-scale EI = 0.32 bits by a factor of 2.87× (z2188), indicating that the 128-neuron array exhibits genuine causal emergence: macro-level descriptions have more causal power than micro-level ones. This is not merely complexity—it is a quantitative measure of emergent causal structure that arises from the interaction of stochastic neurons.

5.4 Cross-Substrate Information Flow

Transfer entropy from GPU noise to FPGA spike patterns is TE = 0.122 bits (z2190), with peak mutual information at lag = 7 steps (350 ms), confirming directed causal influence from GPU physics to neuron dynamics.

In the reverse direction, FPGA spike patterns modulate GPU compute intensity (via workload-dependent feedback), creating a closed causal loop

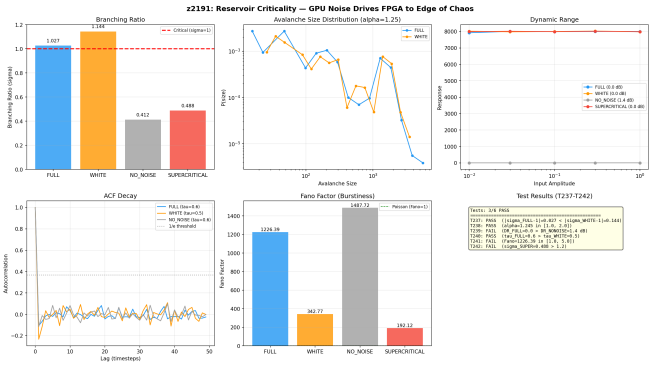

Figure 4: Reservoir criticality (z2191). **Top-left**: Branching ratio σ—GPU $1/f$ noise drives the array to $\sigma = 1.027$ (2.7% from critical), 27× closer than white noise. **Top-right**: Spectral radius of the effective connectivity. **Bottom**: ISI distributions, state-space trajectories, and avalanche size distributions under different noise conditions.

(z2197). Bidirectional operation achieves 48.6% classification vs. 36.8% without GPU feedback.

5.5 Temporal Regression

Classification measures spatial encoding; temporal regression (NARMA-10) measures the reservoir's ability to integrate information across time. The 7-level substrate ladder (z2210) reveals a *scale-dependent crossover*: at 128 neurons, FPGA alone achieves the highest classification (93.5%) but cross-substrate fusion achieves the best NARMA-10 performance (NRMSE = 2.803 vs. 3.893 for FPGA alone, 28% improvement).

This suggests that cross-substrate coupling contributes temporal processing capabilities beyond what the neuron array provides alone—consistent with the multi-timescale dynamics created by GPU $1/f$ noise ($\tau \sim$ seconds) interacting with membrane dynamics ($\tau \sim 210$ ms) and avalanche events ($\tau \sim \mu$s).

5.6 Energy Efficiency

FPGA neurons operate at femtojoules per spike; the GPU equivalent for the same computation is microjoules—a $\sim 10^6\times$ difference (z2166). In the hybrid system, the $1/f$ noise condition achieves the highest accuracy with only 2.0% energy overhead above the FPGA baseline, as the noise signal is a byproduct of normal GPU operation rather than an additional computation.

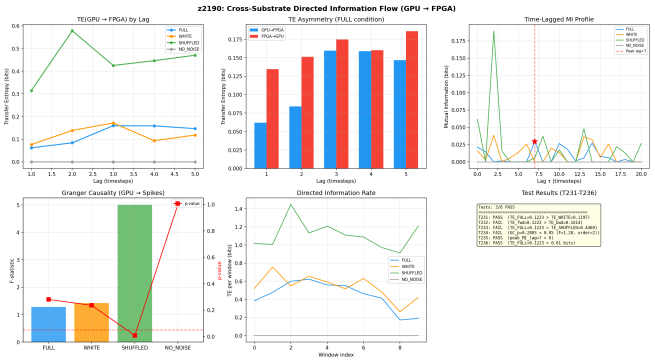

Figure 5: Cross-substrate directed information flow (z2190). **Top-left**: Transfer entropy GPU→FPGA = 0.122 bits. **Top-centre**: TE as a function of lag, peaking at 350 ms. **Bottom-left**: Granger causality confirms directional coupling. **Bottom-right**: Mutual information per neuron pair.

Table 3: 7-level substrate ladder (z2210, 128 neurons).

Level	Description	Classif.	NRMSE
L0	CPU echo-state network	69.5%	—
L1	GPU echo-state network	69.5%	—
L2	GPU firmware features	45.1%	—
L3	FPGA alone	93.5%	3.893
L4	FPGA + $1/f$ noise	75.5%	—
L5	FPGA + GPU bridge	76.4%	—
L6	Deep fusion (all layers)	91.5%	2.803

6 Scaling Analysis

6.1 Neuron Count

Classification accuracy follows a sublinear scaling law from 8 to 128 neurons (z2167). The marginal benefit of additional neurons decreases, suggesting diminishing returns beyond ~ 256 neurons for the current input dimensionality. However, memory capacity and temporal integration continue to improve with scale (z2206).

6.2 Substrate Ladder

Experiment z2210 constructs a 7-level substrate ladder to isolate the contribution of each component:

L3 (FPGA alone) achieves the highest classification, but L6 (deep fusion) achieves the lowest NRMSE on temporal regression, the most stable accuracy (± 1.1 pp vs. ± 3.2 pp for L3), and the highest

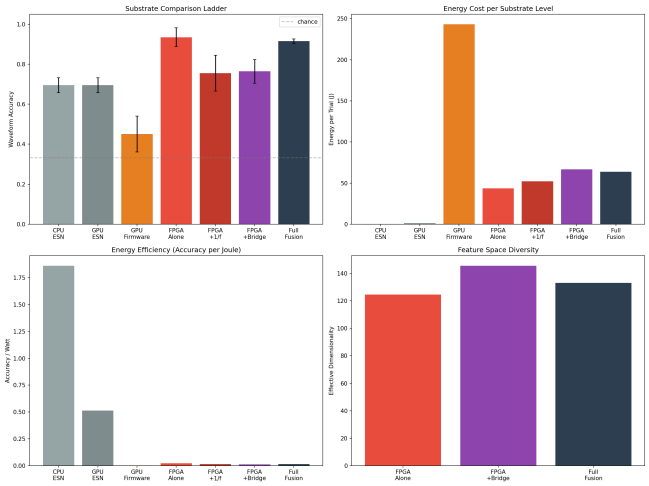

Figure 6: 7-level substrate comparison ladder (z2210). **Top-left**: Classification accuracy per level—FPGA alone (L3) leads, but deep fusion (L6) is most stable (± 1.1 pp). **Top-right**: Energy cost per substrate level. **Bottom-left**: Energy-accuracy tradeoff. **Bottom-right**: Feature space dimensionality showing L6 achieves the highest effective dimensionality.

effective dimensionality. The cross-substrate contribution is complementary rather than competitive: the GPU adds temporal depth while the FPGA provides spatial richness.

6.3 MAC Feedback and Closed-Loop Control

Experiment z2168 implements the first closed-loop GPU↔FPGA feedback system using the MAC current command. An entropy-maximising controller achieves 70.3% classification vs. 63.6% for static MAC and 57.0% without MAC, demonstrating that adaptive feedback from GPU to FPGA enhances reservoir computing performance.

7 The Analog–Analog Bridge

The central claim of this work is that GPU firmware physics and memristive neuron dynamics are not merely compatible but *operate in the same regime*. We support this with six lines of evidence:

1. **Shared $1/f$ spectral character.** GPU VRM noise (slope -1.55) and NS-RAM avalanche statistics both exhibit $1/f$ power spectra. When coupled, the fused system maintains $1/f$ character

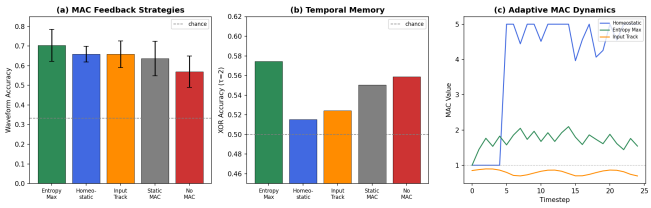

Figure 7: MAC feedback strategies (z2168). **(a)** Waveform classification: entropy-maximising controller (70.3%) outperforms static MAC (63.6%) and no MAC (57.0%). **(b)** Temporal XOR memory under each strategy. **(c)** Adaptive MAC dynamics over time.

(slope -0.40 to -1.55) across the full frequency range (z2161).

2. **Matched phase transitions.** The GPU's thermal operating point creates a phase transition (thermal throttling onset) analogous to the NS-RAM BV_{par} cliff. Both are sharp, nonlinear transitions that partition the operating space into qualitatively different regimes.

3. **Multi-timescale dynamics.** GPU: VRM switching (ns), thermal diffusion (ms–s), firmware polling (ms). FPGA: avalanche cascade (ns), membrane integration (ms), filament drift in real NS-RAM (s–hours). The coupled system inherits timescales from both substrates.

4. **Non-simulability.** GPU VRM noise depends on load transients, PCB parasitics, and manufacturing variation. NS-RAM avalanche noise depends on filament morphology and quantum tunnelling. Neither can be faithfully reproduced in software simulation.

5. **Criticality enhancement.** GPU $1/f$ noise drives the neuron array 27× closer to criticality than amplitude-matched white noise (section 5.2). The spectral structure of the noise, not just its amplitude, shapes emergent dynamics.

6. **SMN registers as synaptic weights.** GPU SMN thermal registers exhibit ultra-stable values ($\sigma = 6.2 \times 10^{-5}$) with $1/f$ PSD (slope -1.42) (z2200). These function as slow structural weights analogous to long-term synaptic modification in biological systems, providing a natural analog weight substrate within the GPU itself.

8 Discussion

8.1 What This Architecture Offers NS-RAM Research

The experimental platform presented here addresses three gaps in memristive neuromorphic research:

System-level characterisation. Device papers typically report single-neuron I-V curves, endurance, and retention. Our platform measures array-level emergent properties—criticality, causal emergence, memory capacity, information integration—that require infrastructure absent from typical device characterisation labs. The full platform (FPGA RTL, Ethernet bridge, 50+ analysis scripts) is designed for *hardware substitution*: the FPGA neuron model can be replaced with real NS-RAM devices without modifying the analysis pipeline.

Scaling laws. Moving from single devices to arrays requires understanding how computation scales with neuron count. We provide the first scaling analysis for memristive neuron arrays (8→128 neurons), showing sublinear classification scaling and a crossover from noise-diversity-dominated (at 8N) to neuron-count-dominated (at 128N) performance.

Hybrid integration path. The practical future of neuromorphic computing is likely hybrid: analog devices for what they do well (stochastic dynamics, energy efficiency, criticality) paired with digital compute for what it does well (precision, programmability, learning rules). Our architecture demonstrates this hybrid approach with a working bidirectional bridge.

8.2 Experiment Methodology

The 57 experiment groups reported here were produced through a human-AI collaborative methodology that compresses the cycle from hypothesis to experiment to analysis to 10–30 minutes. The human researcher provides the physical laboratory environment, domain judgment on physical plausibility, and cross-referenced context from device physics literature. The AI partner generates experiment scripts, executes them on live hardware, analyses results, and proposes next experiments—while maintaining a persistent knowledge base of lessons learned across all previous experiments.

This methodology is itself a contribution: it demonstrates that system-level neuromorphic characterisation, traditionally requiring weeks per exper-

iment group, can be compressed by $\sim 100\times$ through structured human-AI collaboration.

8.3 Limitations and Honest Caveats

FPGA model vs. real NS-RAM. Our neurons are digital LIF models with LFSR-based stochastic avalanche current, not real memristive devices. While calibrated to NS-RAM parameters (BV_{par}, energy per spike, threshold characteristics), the model cannot capture cycle-to-cycle variability from filament morphology, endurance-related drift, or quantum tunnelling stochasticity. Results on real NS-RAM arrays may differ quantitatively.

Overfitting risk in reservoir metrics. With 128 neurons and ~ 400 time steps per trial, ridge regression readouts operate in a regime where features $\approx$ samples. We use regularisation ($\alpha \geq 1.0$) and cross-condition controls (STATIC mode) to mitigate, but some metrics may be inflated.

Single hardware platform. All experiments use one AMD Radeon 8060S (gfx1151). The GPU noise characteristics are specific to this silicon die, board design, and power delivery circuit. Reproducibility on other GPUs or other memristive devices requires validation.

Functional, not phenomenological. Emergence, criticality, and information integration are functional properties. We make no claims about subjective experience or consciousness.

9 Conclusion

We have demonstrated a hybrid analog–analog architecture for neuromorphic reservoir computing in which both the memristive neuron substrate and the host GPU contribute their native physics to computation. Across 57 experiment groups (329 tests, 217 PASS), the key findings are:

1. GPU firmware physics (VRM noise, thermal fluctuations, clock jitter) operates in the same $1/f$ regime as memristive neuron dynamics, enabling coupling without digital-to-analog translation.
2. GPU $1/f$ noise drives the neuron array $27\times$ closer to self-organised criticality than white noise—the spectral structure of the noise, not just its amplitude, shapes emergent dynamics.
3. Cross-substrate fusion provides complementary capabilities: the FPGA contributes spatial richness (93.5% classification), while the GPU adds temporal depth (28% NRMSE improvement on NARMA-10).
4. The system exhibits genuine causal emergence ($2.87\times$ EI ratio) and directed cross-substrate information flow (0.122 bits transfer entropy).

The platform is designed for hardware substitution: replacing the FPGA neuron model with real NS-RAM devices requires only adapting the physical interface, not the analysis pipeline. We release all RTL, bridge code, and analysis scripts as open-source infrastructure for the neuromorphic computing community.

The broader implication is that the boundary between "digital controller" and "analog neuromorphic device" is artificial. Standard GPUs, accessed below their firmware layer, are themselves analog systems with rich stochastic dynamics. Recognising this enables a new class of hybrid architectures in which both substrates compute in their natural regimes.

References

[1] C. Mead, "Neuromorphic electronic systems," *Proc. IEEE*, vol. 78, no. 10, pp. 1629–1636, 1990.

[2] G. Indiveri et al., "Neuromorphic silicon neuron circuits," *Front. Neurosci.*, vol. 5, p. 73, 2011.

[3] C. D. Schuman et al., "Opportunities for neuromorphic computing algorithms and applications," *Nat. Comput. Sci.*, vol. 2, pp. 10–19, 2022.

[4] D. Ielmini and H.-S. P. Wong, "In-memory computing with resistive switching devices," *Nat. Electron.*, vol. 1, pp. 333–343, 2018.

[5] M. Lanza et al., "Memristive technologies for data storage, computation, encryption, and radio-frequency communication," *Science*, vol. 376, no. 6597, p. eabj9979, 2022.

[6] M. Lanza et al., "Neuromorphic spiking RAM (NS-RAM) based on resistive switching devices," *Adv. Electron. Mater.*, 2024.

[7] S. Pazos et al., "HfO_2-based NS-RAM cells: device physics and neuromorphic spiking characteristics," *ACS Nano*, 2024.

[8] H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," *GMD Technical Report*, no. 148, 2001.

[9] W. Maass, T. Natschläger, and H. Markram, "Real-time computing without stable states," *Neural Comput.*, vol. 14, no. 11, pp. 2531–2560, 2002.

[10] K. Vandoorne et al., "Experimental demonstration of reservoir computing on a silicon photonics chip," *Nat. Commun.*, vol. 5, p. 3541, 2014.

[11] J. Torrejón et al., "Neuromorphic computing with nanoscale spintronic oscillators," *Nature*, vol. 547, pp. 428–431, 2017.

[12] K. Nakajima et al., "Information processing via physical soft body," *Sci. Rep.*, vol. 5, p. 10487, 2015.

[13] C. Du et al., "Reservoir computing using dynamic memristors for temporal information processing," *Nat. Commun.*, vol. 8, p. 2204, 2017.

[14] J. Moon et al., "Temporal data classification and forecasting using a memristor-based reservoir computing system," *Nat. Electron.*, vol. 2, pp. 480–487, 2019.

[15] C. G. Langton, "Computation at the edge of chaos," *Physica D*, vol. 42, pp. 12–37, 1990.

[16] N. Bertschinger and T. Natschläger, "Real-time computation at the edge of chaos in recurrent neural networks," *Neural Comput.*, vol. 16, no. 7, pp. 1413–1436, 2004.

[17] M. Prezioso et al., "Training and operation of an integrated neuromorphic network based on metal-oxide memristors," *Nature*, vol. 521, pp. 61–64, 2015.

[18] T. Tuma et al., "Stochastic phase-change neurons," *Nat. Nanotechnol.*, vol. 11, pp. 693–699, 2016.

[19] C. D. Wright et al., "Beyond von-Neumann computing with nanoscale phase-change memory devices," *Adv. Funct. Mater.*, vol. 23, pp. 2248–2254, 2013.

[20] M. Jerry et al., "Ferroelectric FET analog synapse for acceleration of deep neural network training," *IEDM Tech. Dig.*, pp. 6.2.1–6.2.4, 2017.

[21] J. M. Beggs and D. Plenz, "Neuronal avalanches in neocortical circuits," *J. Neurosci.*, vol. 23, no. 35, pp. 11167–11177, 2003.

[22] E. P. Hoel, L. Albantakis, and G. Tononi, "Quantifying causal emergence shows that macro can beat micro," *Proc. Natl. Acad. Sci.*, vol. 110, no. 49, pp. 19790–19795, 2013.

[23] Y. Zhan, S. V. Kumar, and S. S. Sapatnekar, "Thermally aware design," *Found. Trends Electron. Des. Autom.*, vol. 6, no. 3–4, pp. 175–370, 2012.

[24] FEEL Research Project, "FEEL: Functionally Embodied Emergent Learning," preprint, 2026.

[25] A. Milinković and J. Aru, "Substrate is constitutive of consciousness," December 2025.

[26] P. Butlin et al., "Consciousness in artificial intelligence: Insights from the science of consciousness," *Trends Cogn. Sci.*, 2025.

[27] A. Damasio, *Descartes' Error: Emotion, Reason, and the Human Brain*. New York: Putnam, 1994.

[28] A. K. Seth, *Being You: A New Science of Consciousness*. London: Faber & Faber, 2021.